

Flash Flood Prediction & Catchment Intelligence System

Master Technical Guide: Overall Understanding, Technology Stack, and End-to-End Operations

Target Region: Mandakini River Catchment & Rudraprayag, Uttarakhand	System Core: AI Forecasting + GIS Mapping + LoRa Mesh + PWA
Backend Validation: 74/74 Pytest Unit & Integration Tests Passing (100%)	Frontend Status: TypeScript Strict Mode Clean (0 Build Errors)
Target Audience: Disaster Authorities, NDRF Responders, Evaluators, Engineers	Published Date: September 17, 2026 - 16:19 IST

1. Overall Understanding View (The 30,000-Foot Blueprint)

What is this system? It is an intelligent, hyper-local, multi-tier disaster early warning and emergency response decision support platform specifically engineered for mountainous Himalayan valleys prone to sudden cloudbursts, debris flows, and catastrophic flash floods (e.g., Kedarnath, Mandakini Basin).

Why is it groundbreaking? Conventional weather forecasts operate on 10–25 km grid resolutions and only issue warnings hours after rain has already started. In steep terrain (average slope > 24°), water runs off into narrow river gorges in **under 30 to 60 minutes**. This platform solves that fatal latency by fusing: (1) **Hyper-local 48-node IoT sensor networks**, (2) **AI-driven 30-120 minute hydrograph forecasting**, (3) **Autonomous LoRaWAN Edge Sirens that trigger even during power/cell tower blackouts**, (4) **Hazard-aware evacuation routing**, and (5) **Offline-first PWA crowdsourcing** for field volunteers.

HIGH-LEVEL 4-TIER SYSTEM ARCHITECTURE BLUEPRINT

- [TIER 1: PHYSICAL TELEMETRY] 48 IoT Gauge Stations (Rainfall mm/h, River Level m, Soil Saturation %, Barometer) + 4 LoRa Gateways
↓ (RS485 / LoRaWAN Mesh / REST Ingestion)
- [TIER 2: INTELLIGENT BACKEND] FastAPI (Async Core) + AI Forecasting Regressors + Explainable Risk Engine + WebSocket Hub
↓ (SQLAlchemy 2.0 ORM + Dual Storage: PostgreSQL PostGIS / Resilient SQLite)
- [TIER 3: AUTOMATED DISASTER DISPATCH] Autonomous LoRa Edge Sirens + SMS/Push Alerts + Dynamic Evacuation Router
↓ (Sub-second Reactive Push)
- [TIER 4: UNIFIED COMMAND UI] React 18 + Leaflet GIS + Recharts + Offline PWA (IndexedDB Queue + Service Worker)

2. Complete Technology Stack Matrix

Layer	Technology / Framework	Specific Modules & Libraries	Role & Why It Was Chosen
Frontend Core	React 18 + TypeScript	Vite 6, React DOM, TS Strict Mode	Type safety across all telemetry models, ultra-fast Vite HMR build, component modularity.
Styling & UI	Tailwind CSS + Custom Tokens	Lucide React Icons, CSS Variables	Clean Royal Blue & White high-contrast design system optimized for emergency command operations.
GIS & Mapping	Leaflet + React-Leaflet	Leaflet 1.9, OpenStreetMap Vector	Dynamic GeoJSON layers for gauge nodes, danger inundation zones, catchment polygons & safe routes.
Data Viz	Recharts + D3 Scales	AreaChart, LineChart, ResponsiveBox	Live multi-horizon hydrographs, rainfall accumulation curves, sensor uptime & prediction error metrics.
PWA / Offline	Vite-Plugin-PWA + Workbox	ServiceWorker, IndexedDB, CacheAPI	App shell caching, background sync queue, and local IndexedDB offline citizen reporting storage.
Backend API	FastAPI (Python 3.12+)	Starlette, Pydantic v2, Uvicorn	Asynchronous high-throughput ASGI framework, automatic schema validation, sub-millisecond serialization.
WebSockets	FastAPI WebSocket Manager	Asyncio, ws_manager (/ws, /ws/sensors)	Real-time bidirectional telemetry streaming and instant siren broadcast pushing to all clients.
Database	PostgreSQL 16 / SQLite	SQLAlchemy 2.0 ORM, Alembic	Resilient dual-mode engine: PostgreSQL/PostGIS in production with zero-config local SQLite fallback.
AI / ML Model	Scikit-Learn + NumPy	GradientBoostingRegressor, RandomForest	Hydrological runoff prediction at t+30m to t+120m, Antecedent Moisture Condition (AMC) indexing.
Security / Auth	OAuth2 Bearer + Passlib	python-jose (JWT), bcrypt hashing	HMAC-SHA256 tokens, 5-tier Role-Based Access Control (Admin, Officer, Responder, Researcher, Citizen).
Edge Simulation	LoRaWAN Mesh Simulator	Asyncio, Edge State Machine	Simulates 48 LoRa nodes, packet loss resilience, autonomous siren broadcast triggers.

3. How the System Works — Step-by-Step Data Pipelines

The system operates across 6 synchronized processing pipelines that turn raw raindrops into life-saving evacuations:

Pipeline 1: Telemetry Ingestion & Real-Time Distribution

- **Edge Gauges:** 48 IoT sensor nodes across 8 Himalayan watersheds measure Rainfall Intensity (mm/h), River Gauge Level (m), Soil Saturation (%), and Temperature.
- **Ingestion:** Ingested via POST /api/sensors/ingest or LoRaWAN gateways. Validated by Pydantic schemas and written synchronously to database.
- **WebSocket Push:** ConnectionManager pushes updates to all active WebSockets (/ws/sensors), updating client charts and map markers in sub-50ms.

Pipeline 2: AI Forecasting Engine & Explainable Risk Scoring

- **Multi-Horizon Predictions:** Gradient Boosted model forecasts future river levels for **t+30m**, **t+60m**, **t+90m**, and **t+120m**.
- **Explainable Risk Engine:** Calculates a composite 0–100 Risk Score: **Rainfall Rate (35%) + Soil Moisture (25%) + Current Water Level (25%) + Terrain Slope/Elevation (15%)**.
- **Severity Stages:** Maps directly to: **LOW (0–39)**, **MODERATE (40–64)**, **HIGH (65–84)**, and **CRITICAL CLOUDBURST (85–100)**.

Pipeline 3: Autonomous LoRaWAN Edge Mesh Sirens

- **Edge Autonomy:** If river rise rate exceeds **+0.45 m/10min** or rainfall exceeds **60 mm/h**, local edge sirens trigger autonomously without internet.
- **LoRa Mesh Relay:** In case of cell network blackout, packets hop across neighboring edge nodes to municipal relays.

Pipeline 4: Hazard-Aware Safe Evacuation Routing

- **Inundation Hazard Buffers:** Watersheds with score ≥ 65 generate active danger polygons.
- **Routing Algorithm:** Evacuation engine calculates safe paths from user coordinates to the nearest shelter (e.g. Kedarnath Helipad Camp), guaranteeing routes do not cross flooded road segments.
- **Shelter Tracking:** Real-time bed occupancy, medical staff, food supplies, and power backup monitoring.

Pipeline 5: Offline-First Citizen Crowdsourcing (PWA)

- **IndexedDB Local Storage:** Citizen reports submitted in zero-connectivity areas are saved in IndexedDB (PENDING).
- **Service Worker Auto-Sync:** When internet reconnects, Service Worker background sync pushes reports to /api/reports (SYNCED).
- **Responder Verification:** Responders inspect photo evidence and adjust confidence scores (0–100%).

Pipeline 6: Demo Orchestration & Simulation Engine

- **4-Stage Cloudburst Simulation:** Stage 1 (Normal Inflow) → Stage 2 (Heavy Rainfall) → Stage 3 (River Surging) → Stage 4 (Cloudburst Peak & Sirens Active).
- **Live Telemetry Injection:** Injects simulated real-time telemetry spikes directly into WebSockets for emergency drills.

4. Role-Based Access Control (RBAC) & Demo Accounts

Role	User Persona	Allowed Modules & Privileges	Default Demo Account
ADMIN	System Administrator	Full superuser access: Sensor calibration, watershed boundaries, user roles, system config	admin@disaster.gov.in Pass: AdminPass123!
GOV OFFICIAL	District Disaster Authority	Command Center console, multi-channel siren triggers, evacuation & shelter management	official@disaster.gov.in Pass: OfficialPass123!
RESPONSE TEAM	NDRF / SDRF Commander	Live hazard maps, citizen report verification, safe convoy routing, rescue status updates	responder@disaster.gov.in Pass: ResponderPass123!
RESEARCHER	Hydrologist & Climate Scientist	Sensor telemetry inspection, AI predictions & hydrographs, historical analytics export	dr.anita.hydrology@research.ac.i n Pass: password123
CITIZEN	Local Resident / Volunteer	Public risk status, localized siren alerts, crowdsourced report submission, safe shelter map	citizen@disaster.gov.in Pass: CitizenPass123!

5. Complete REST & WebSocket API Directory (40+ Endpoints)

Module	HTTP Method & Path	Description & Response Payload
Health	GET /api/health GET /api/health/db	Health check, system uptime, and PostgreSQL/SQLite connectivity ping.
Auth	POST /api/auth/register POST /api/auth/login GET /api/auth/me	User onboarding, JWT bearer token issuance with bcrypt verification, and user profile.
Overview	GET /api/overview/metrics	Summary KPIs: active sensors, watersheds at risk, active warnings, and critical zones.
Sensors	GET /api/sensors GET /api/sensors/{id}/readings POST /api/sensors/ingest	List sensors with filters, retrieve historical readings, and ingest real-time hardware telemetry.
Watersheds	GET /api/watersheds	List all Himalayan catchment basins (Mandakini, Alaknanda, Bhagirathi, etc.).
GIS Map	GET /api/map/sensors GET /api/map/watersheds GET /api/map/risk-zones GET /api/map/evacuation-centers	Standard GeoJSON FeatureCollections for sensors, watershed boundaries, danger zones, and shelters.
Risk Engine	GET /api/risk/watershed/{id} GET /api/risk/all GET /api/risk/configuration	Returns composite flood risk index (0–100), factor weightings, and threshold settings.
AI Prediction	POST /api/predictions/generate GET /api/predictions/watershed/{id} GET /api/predictions/metadata	Produces t+30m, t+60m, t+90m, t+120m river level forecasts and model accuracy metrics.
Alerts	GET /api/alerts POST /api/alerts/{id}/acknowledge POST /api/alerts/evaluate	Retrieves alerts, evaluates threshold rules across basins, acknowledges alerts, and logs siren events.
Citizen Reports	POST /api/reports GET /api/reports POST /api/reports/{id}/verify	Accepts crowdsourced field observations and provides responder verification workflow.
Evacuation	GET /api/evacuation/centers GET /api/evacuation/nearest GET /api/evacuation/routes	Shelter inventory, nearest shelter search via Haversine, and safe evacuation routing avoiding flood polygons.
Command Center	GET /api/command-center/overview GET /api/command-center/timeline GET /api/command-center/trends	DEFCON status overview, real-time incident event timeline, sensor health, and rainfall trends.
Edge Mesh	GET /api/edge/overview POST /api/edge/simulate-failure POST /api/edge/restore-network	Monitors 48 LoRa nodes, simulates telecom blackouts, and tests edge packet buffering.
Analytics	GET /api/analytics/summary GET /api/analytics/charts	Average/max rainfall, peak river water level, average soil moisture, prediction error metrics (MAE/RMSE).
Demo Mode	GET /api/demo/status POST /api/demo/start POST /api/demo/stop	Controls the 4-stage live flood scenario demo simulation.
WebSockets	WS /ws WS /ws/sensors WS /ws/alerts	High-speed asynchronous WebSocket feeds for live sensor telemetry and instantaneous broadcast alert pushes.

6. Relational Database Schema (12 Core Entities)

- **1. users:** id (PK), name, email (unique), password_hash, role (Enum), phone, district, state, created_at

- **2. watersheds:** id (PK), code (unique), name, river_name, drainage_area_km2, base_flow_m3s, average_slope_deg, elevation_m
- **3. sensors:** id (PK), sensor_code (unique), watershed_id (FK), name, sensor_type, latitude, longitude, status, battery_pct
- **4. sensor_readings:** id (PK), sensor_id (FK), timestamp, rainfall_rate_mmh, water_level_m, soil_moisture_pct, temperature_c
- **5. risk_evaluations:** id (PK), watershed_id (FK), evaluation_time, risk_score (0-100), risk_level, dominant_factor, confidence_pct
- **6. predictions:** id (PK), watershed_id (FK), prediction_time, horizon_minutes (30/60/90/120), predicted_level_m, risk_score
- **7. alerts:** id (PK), watershed_id (FK), severity, status, title, description, triggered_at, acknowledged_at
- **8. alert_notifications:** id (PK), alert_id (FK), channel (SMS/VOICE/LORA_SIREN), recipient, status, sent_at
- **9. citizen_reports:** id (PK), user_id (FK), report_type, description, latitude, longitude, confidence_score, verification_status
- **10. evacuation_centers:** id (PK), name, district, total_capacity, current_occupancy, latitude, longitude, elevation_m, is_active
- **11. evacuation_routes:** id (PK), origin_lat/lng, center_id (FK), distance_km, estimated_minutes, safety_score, road_status
- **12. edge_nodes:** id (PK), node_id (unique), watershed_id (FK), status, battery_level, rssi_dbm, siren_active, packet_queue_count

7. Execution, Testing & Startup Guide

1. Start FastAPI Backend Server (Port 8000)

```
cd backend
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

2. Start Vite React Frontend Application (Port 5173)

```
cd frontend
npm run dev -- --host 0.0.0.0 --port 5173
```

3. Run Backend Automated Test Suite (74 Tests)

```
pytest backend/tests/ -v
```

End of Master Technical Specification — Prepared for Ministry of Jal Shakti, NDMA, and Technical Evaluation Committee.